

Delineation, Systems, and Synthesis

From admissibility to linguistic explanation

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Delineation asks where a modeled system begins and ends

Question

Which variables belong to the linguistic system, and which remain environmental inputs?

- A system is an abstraction from interacting processes.
- Boundaries determine which relations count as internal.
- Open and closed descriptions support different claims about choice and control.

MOLR §6.5; Preface roadmap

Where we are

1. Systems as abstractions
2. Open and closed systems
3. Course synthesis

A system is selected, not simply found

- Components are grouped because specified relations are preserved.
- Inputs and outputs depend on the chosen boundary.
- The same event may be internal to one model and environmental to another.
- A useful boundary supports explanation without pretending isolation.

Guided analysis

Draw two boundaries around a conversation: one grammatical, one social. What crosses each boundary?

Transforms can reveal a useful invariant description

Euler's identity connects several fundamental constants:

$$e^{i\pi} + 1 = 0.$$

A generating function packages a sequence $f(n)$:

$$P(f, x) = \sum_{n \in \mathbb{N}} f(n)x^n.$$

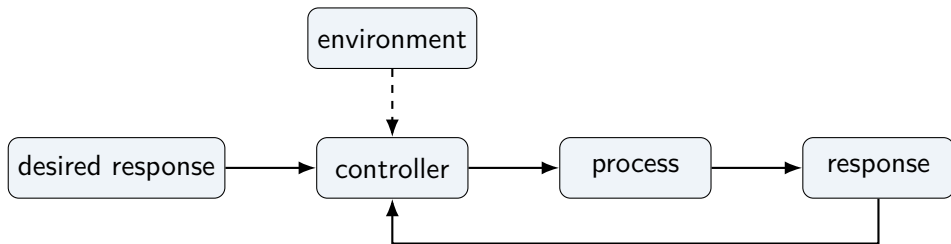
- A transform changes representation to make selected relations tractable.
- The transformed object must still be interpreted against the original problem.

MOLR §6.5

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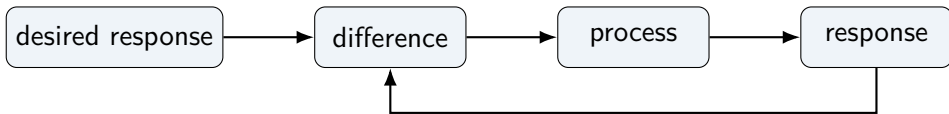
An open system retains an independent environmental input



Take-away

Feedback constrains action, but an open controller can also receive information not reducible to the feedback loop.

A closed loop fixes the route from discrepancy to adjustment



- The model can explain regulation under its fixed policy.
- It cannot by itself explain the selection or revision of that policy.

Choice and decision are not the same abstraction

Decision

Select an action under a fixed policy and specified inputs.

Choice

Select or revise which policy, goal, or system of relevance will govern.

Question

Which aspects of language use can be modeled as decision, and which require choice?

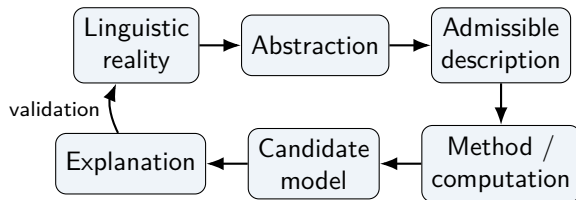
Where we are

1. Systems as abstractions
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The course has moved through three layers

Vocabulary	sets, strings, relations, functions, composition, recursion
Description	admissibility, grammar families, regexps, formal grammars
Method	CKY, automata, parsing, conversion, transduction
Explanation	contingent homomorphism, system boundary, empirical validation

The complete route returns to linguistic reality



Take-away

Every formal route must eventually close through empirical validation; otherwise it remains a possible mathematics.

The central caution is also the central opportunity

- A formalism makes consequences inspectable.
- It also makes its selected invariants and suppressed variants inspectable.
- Comparison across languages, acquisition stages, and activities tests whether the selection survives.
- Mathematics contributes most when it sharpens the route back to evidence.

Question

Which abstraction from this course changed how you see a familiar linguistic analysis?

Third hour: rebuild the roadmap

In groups, choose one phenomenon from morphology, phonology, syntax, or semantics.

- ① State the linguistic reality to be explained.
- ② Select an invariant and identify its variants.
- ③ Give an admissibility description.
- ④ Choose a computational method only if one is needed.
- ⑤ State the proposed correspondence and validation evidence.
- ⑥ Redraw the system boundary and note what changes.

Week 12 synthesis

- ① System boundaries are explanatory choices.
- ② Open and closed systems retain different sources of determination.
- ③ Transforms aid analysis but do not interpret themselves.
- ④ Description, computation, modeling, and explanation form distinct stages.
- ⑤ The final test is correspondence with linguistic reality.

Question

Next week: reflection and final assessment. No new formal vocabulary.